

Initial Labor Market Conditions: Unequal Effects by Parental Income

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This paper examines the unequal impact of initial labor market conditions by parental income and their role in shaping long-term labor market outcomes. Using large-scale Canadian administrative tax data, I analyze the relationship between parental income, aggregate economic conditions, and wage dynamics for new labor market entrants. I find that entering the labor market during an economic downturn leads to significant and persistent earnings losses, with disproportionately larger effects for individuals from low-income families. A 1 p.p. higher initial unemployment rate reduces annual earnings by approximately \$2,000 real CAD for all workers over the first decade after graduation, with earnings declining 15 percent more for those from families in the bottom 20% of the income distribution. To explore the underlying mechanisms, I develop a continuous time heterogeneous agent model with financial constraints and a frictional labor market. The model highlights how parental income can act as a safety net: individuals from low-income families are closer to their borrowing constraints and more likely to enter jobs with lower earnings, particularly in the presence of larger search frictions. Recessions amplify this mechanism by reducing job-finding rates in equilibrium, further limiting access to high paying jobs. Counterfactual simulations suggest policies such as targeted grants for graduates entering the labor market could mitigate scarring effects and improve outcomes for disadvantaged workers.

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1. Introduction

This paper examines the persistence of intergenerational income inequality, focusing on the interplay between parental income and initial labor market conditions. Parents can provide insurance against idiosyncratic income risks during their children's lifetimes, serving as a financial safety net for their adult children. For financially constrained individuals entering the labor market, this safety net can be particularly crucial to insulate against economic shocks.

At the same time, the conditions under which individuals enter the labor market have long-lasting effects. College students who graduate and enter the labor market in a recession experience average cumulative lifetime earnings losses of 5 percent and decreased wages for up to 10 years (Oreopoulos et al. 2012). Existing research largely treats these initial labor market effects as uniform across individuals, overlooking heterogeneity by socioeconomic status. What is still not well understood is how the effects of initial labor market conditions differ across the parental income distribution. This paper addresses this gap by investigating how initial labor market conditions shape the long-term earnings and employment outcomes across the parental income distribution.

I introduce and establish evidence for a parental insurance mechanism, whereby individuals from lower-income families find themselves closer to the borrowing constraint and more likely to be liquidity constrained compared to their peers with better-off parents. These financial constraints generate different values of unemployment and hence different outside options during the process of labor market search. This heterogeneity initially depends on parental income, and in turn makes individuals with lower-income parents more likely to go into jobs with lower earnings. When initial labor market conditions worsen, such as during recessions, this mechanism is further intensified. Declining job-finding rates and longer unemployment spells further depress outside options, leading to lower earnings in subsequent jobs, especially for individuals from low-income backgrounds.

To analyze this mechanism, this paper uses a rich dataset of Canadian administrative records that links tax files, post-secondary education records and parental income tax data. This dataset allows me to track individuals across their early careers with a high degree of precision and measure how initial labor market conditions affect earnings and employment outcomes across the parental income distribution. I find that initial labor market conditions affect entrants across the parental income distribution, but these effects are unequal. I document that when initial labor market conditions are worse and the initial unemployment rate rises by 1 percentage point, yearly earnings are on average 10% lower than trend, and these lowered earnings persist for 8 years after graduation.

One of this paper's main contributions is documenting the unequal effects by parental income. I find that labor earnings decline 5 percentage points more for entrants from the poorest 20% of families. These effects are concentrated in the first three years after graduation after which individuals from all family backgrounds suffer similarly lowered earnings for eight years after graduation. The differential effects by parental income are robust to conditioning on a wide range of observables, including gender, degree, major, geography, industry group, institution quality and post-secondary institution fixed effects, with low-income individuals continuing to be especially vulnerable to initial labor market conditions.

Using these empirical findings, I build a heterogeneous-agent model with search frictions, different initial assets and human capital. The core of the model is the interaction between incomplete asset markets and frictional labor search. Individuals are endowed with heterogeneous human capital and initial assets, capturing differences in skill by schooling and different parental safety-nets proportional to parental income. Agents are unable to insure themselves against unemployment shocks - when jobless, they can only dissave to smooth consumption. Unemployed workers frictionally meet vacancies and produce with a technology combining human and physical capital. Upon meeting, the workers and firms bargain over wages, which depend crucially on the worker's outside option in unemployment and therefore current assets. Lower asset holdings, for instance due

to a weaker parental safety net, reduce the worker's bargaining position and make the acceptance of lower wages more likely. This mechanism thus maps inequality in the parental safety net into inequalities in labor-market earnings, consistent with the empirical evidence. I use the model to quantify the importance of initial assets and human capital and to assess their roles in amplifying or mitigating the effects of initial labor market conditions.

My primary contribution lies in being among the first to document disparities in earnings outcomes for graduates entering the labor market during a recession, based on parental income groups. These findings build on an empirical literature studying the effects of graduating in a recession. Additionally, I contribute by developing and solving a quantitative model, offering a theoretical framework to explain and isolate the economic mechanism as well as simulate policies, which complements the predominantly empirical focus of existing research.

Related Literature: The literature finds graduating in a recession has a lasting negative impact on wages. Oreopoulos, von Wachter and Heisz (2012) document that a recession at labor market entry leads to lower labor earnings for up to the next 10 years. The authors examine male college graduates in Canada from 1982 to 1999, and find that graduating in a recession amounts to a decrease in about 8% of cumulative lifetime earnings. Kahn (2010) finds similar persistently lower wages in male college graduates in the US from 1979 to 1989 using the NLSY79. Similar patterns are found by Genda et al. (2010) and Speer (2015) among male high school graduates. The initial negative impact for high school graduates is larger than that for college graduates, and therefore the income losses are more substantial. However, little is known about the differential effects of labor market entry in a recession across the parental income distribution.

To the best of my knowledge, no paper studies the effects of parental income on graduating and entering the labor market in a recession. A growing empirical literature documents the persistence of intergenerational income inequality and the role of parental income in influencing children's lifetime earnings, including Chadwick and Solong (2002) and

Chetty et. al. (2014) in the United States and Corak (2013) in Canada, as well as the role of post-secondary education (see for example Torche, 2011 and Chetty et. al., 2017). The inter-generational relationship of employers between parents and children has recently been documented empirically (Stinson and Wignall, 2014; Kaila et. al., 2024) and is increasing in paternal earnings (Corak and Piraino, 2011), which is expected to widen inequality by family income.

More broadly, family insurance has long been a topic of interest. The literature has demonstrated that parents do insure children by making transfers to less well-off children, as found by Cox (1990), McGarry (2016) and Ameriks et al. (2016). Choi, McGarry and Schoeni (2016) find that extended family income affects individuals' own consumption, and Boar (2021) shows that parents accumulate precautionary savings to insure their children against labor income risk. Moreover, Kaplan (2012) demonstrates moving into the parental home is an important channel to insure against labor market risk, focusing on the effects for male high-school graduates.

The paper is organized as follows. Section 2 describes the data and empirical results, section 3 presents the quantitative model, and section 4 concludes.

2. Empirical Section

We begin by presenting several novel empirical findings. This section first introduces the administrative tax data and defines parental income as used throughout the paper. I then examine the relationship between earnings and parental income, starting with aggregate cohort-level trends, followed by regression analysis on the microdata.

2.1. Data Description

This paper uses Canadian administrative data that combines a panel of individual tax files linked to the tax files of family members, together with a database of post-secondary students in Canada (the T1FF-PSIS). My sample period focuses on students in the years

2006 to 2020. I use the annual unemployment rate at graduation to characterize the initial labor market conditions, following the literature in Oreopoulos et al. (2012). Figure A1 in the appendix shows the variation in the unemployment rate in Canada over time, with recession periods shaded in light gray. The main recession in my sample period is the Great Recession of 2008-09, during which the national unemployment rate for workers aged 15 and up increased by approximately 2 percentage points. In 2015, an oil shock dampened GDP and moderately increased the unemployment rate into 2016, though the economic contraction was not formally classified as a recession. I exclude graduates and new labor market entrants during the Covid recession in 2020 due to the lack of available data in the years following the recession.

2.2. Measure of Parental Income

This paper measures parental income by using the mean parental pre-tax income for each individual observed in the dataset. All income variables are inflation adjusted. This measure of parental income therefore incorporates not only parental labor income, but also income including capital gains, returns on interest and business income, making it a more holistic measure of total parental income and overall wealth. Taking the average parental income over several years also helps ensure we are measuring a good proxy of permanent parental income that is less influenced by temporary income fluctuations. Throughout the paper, I show results by parental income classified into two groups, those in the bottom 20% and top 80% of average real parental income, where the quintile categories are CPI adjusted and defined across all years. My analysis will compare individuals with different groups of parental income who have the same graduation year. I follow the outcomes of these individuals over years after graduation. The intergenerational income inequality literature commonly analyses parental income in quintiles, with specific interest on individuals from the bottom income quintile. The bottom 20% group also roughly corresponds to poor and liquidity constrained individuals, who Kaplan and Violante (2014) refer to as the poor hand-to-mouth. In robustness checks, I also run my analysis using groups defined on mean parental income before graduation, five quintile groups as well

as using continuous parental income. The results are largely similar.

2.3. Cohort Level Results

I begin by presenting aggregate labor outcomes for graduation cohorts, analyzing the relationship between parental income and the unemployment rate (UR) at the time of graduation. Figure 1 below shows average labor earnings by group in real Canadian dollars, on the y-axis, plotted against years after graduation on the x-axis. This figure presents raw group means by graduation cohort, parental income group, and years after graduation without conditioning on covariates and includes individuals with zero earnings. The year of graduation is denoted at time zero with a dashed line. Each subplot corresponds to a parental income group. In the figure below, we can see individuals whose parents are in the bottom 20% of parental pre-tax income in the left subplot, and individuals whose parents are in the top 80% of the parental income distribution in the right subplot. Each line denotes a graduating cohort that is colored according to the annual total unemployment rate at graduation, with light blue representing graduating into a good labor market with low unemployment, and orange denoting a more difficult initial labor market with a high unemployment rate.

We observe persistent and long-lasting differences in labor earnings by initial aggregate conditions. For all levels of parental income, Figure 1 shows that individuals who graduate and enter the labor market in better initial conditions, in blue, have higher earnings on average for approximately eight to ten years compared to those who graduate into a worse labor market. This finding is in line with Oreopoulos et. al (2012), who document lasting scarring effects for male college graduates in the 1980s and 1990s. The persistence of initial labor market conditions is surprising, as the effects on mean wages far outlast the duration of the recession, which ended in 2009, and wage effects remain even after elevated unemployment rates subside by 2013.

Moreover, we can also see differential effects of earnings across the parental income distribution. In Figure 1, we observe that labor earnings have different profiles by parental

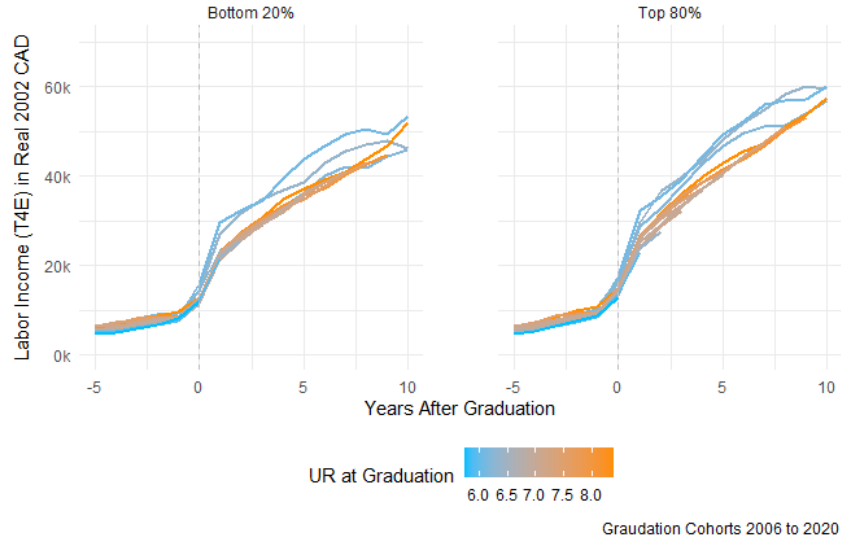


FIGURE 1. Labor Earnings by Parental Income Group and Initial Labor Market Conditions

income groups irrespective of initial labor market conditions, with individuals from low-income backgrounds having both a lower level and a flatter slope of earnings on average. This effect is not immediate, since Figure 1 visualizes only labor earnings, which excludes potential transfers from richer parents to their children. It is therefore not obvious how labor earnings would be directly affected by parental income group. I take these average differences into account in the micro level regressions in the following section.

2.4. Regression Specification

To further understand the relationship between initial labor market conditions and parental income, I analyze the microdata via regression analysis. I focus on individuals with at least three years of non-missing parental income to proxy for permanent income and, to the best of my ability, and reduce effects of temporary parental income fluctuations. I concentrate on individuals who are aged 26 and younger at the time of their first graduation as they are less likely to have extensive work experience prior to college. My sample contains approximately 200 thousand unique individuals and 1.2 million individuals over time who graduated from post-secondary in the sample period. Post-secondary education is defined as education following secondary school, and therefore

after high school. It includes individuals who complete one to two year associate degrees and certificates at colleges, as well as those who finish bachelor's degrees and advanced degrees including masters' degrees.

I run the following regression specification dynamically over years after graduation, denoted by $j = t - c$, where c is the graduation year and t is the calendar year:

$$\begin{aligned}
 y_{ctpi} = & \alpha + \underbrace{\sum_{j=0}^{10} \beta_{p,j=(t-c)} \text{Parental Inc Group} \times \mathbb{1}_{j=t-c}}_{\text{Average effect of parental income group on indiv earnings}} \\
 & + \underbrace{\sum_{j=0}^{10} \gamma_{0,j=(t-c)} UR_{c0} \times \mathbb{1}_{j=t-c}}_{\text{Average effect of initial UR rate}} + X'_i \delta + \phi_t \\
 & + \underbrace{\sum_{j=0}^{10} \gamma_{p,j=(t-c)} UR_{c0} \times \text{Parental Inc Group} \times \mathbb{1}_{j=t-c}}_{\text{Additional effect of having parental income group } p} + u_{ctpi}
 \end{aligned} \tag{1}$$

In my main specification, the dependent variable y_{ctpi} denotes the individual i 's real labor earnings in CPI adjusted Canadian dollars, including zeros if the individual has no labor earnings. I denote the graduation cohort year by c , calendar year t and parental income group p . In some specifications, the dependent variable is restricted to positive labor earnings, and I can therefore look at log earnings. I run further robustness with positive labor earnings above ten thousand 2002 real Canadian dollars, which approximates one year of full-time minimum wage earnings throughout my sample. The effect over time is denoted by $j = (t - c)$, the number of years after graduation.

My main variables of interest are the mean effect of parental income on earnings over time, captured by $\beta_{p,j}$, and the average effect of the initial unemployment rate on earnings in the years following graduation by parental income group p . The average effect of the initial unemployment rate at graduation UR_{c0} , is denoted by $\gamma_{0,j}$ and captures the effect

for the reference parental income group, which is the group of parents in the bottom 20% of the parental income distribution. The interaction term $\gamma_{p,j}$ is the additional affect of the initial unemployment rate at graduation for individuals whose parents are in the other parental income groups in the top 80% of the parental income distribution. This interaction term $\gamma_{p,j}$ is a key parameter of interest; if it is significantly different than zero, then there is evidence of that initial labor market conditions affect entrants differently depending on parental income. With the reference group being those from the lowest socio-economic background, a negative interaction term $\gamma_{p,j} < 0$ signifies that individuals with richer parents have lower earnings when the unemployment rate at graduation UR_{c0} is one percentage point higher. Conversely, if the interaction term is positive, $\gamma_{p,j} > 0$, individuals from poorer family backgrounds fare worse and have lower earnings than their peers from richer backgrounds when the initial unemployment rate increases.

The regression specification conditions on a number of co-variates. In the baseline specification, I condition on the individual's current age as well as year fixed effects. I progressively condition on additional covariates including gender, currently being enrolled in post-secondary, the degree type, and broad major grouping. I also use information on the individual's current principal industry of employment at the two-digit NAICS level. Degree type is classified by the International Standard Classification of Education (ISCED), which captures differences in post-secondary degrees including between a two-year degree, bachelors' degree or equivalent, masters' degree etc. Major groupings in the PSIS has several available variables categorizing field of study by the Classification of Instructional Programs (CIP) Canada 2016 version, including a primary grouping code (CIPPG), a 2-digit code, and a 4-digit code. I condition on the primary grouping code (CIPPG) in the main results, and have run robustness checks using more detailed 2-digit and 4-digit CIP major codes.

2.5. Regression Results

I find that individuals who graduate with a 1 percentage point higher unemployment rate have persistent earnings declines of approximately \$1700 real 2002 CAD per year, equivalent to nearly \$2700 in 2024 Canadian dollars. This result is robust to numerous control variables including age, gender, degree, type of institution, and major grouping. Moreover, the effect of initial labor market conditions on the path of earnings is highly persistent, with earnings being on average \$1700 CAD lower per year for 10 years after graduation, when the unemployment rate at graduation is 1 percentage point higher. That is to say, when comparing two individuals with the same set of observables who are in the same parental income group, those who graduate from post-secondary in a year with a worse labor market and higher unemployment rate earn on average CAD \$1700 less labor earnings per year for ten years. This effect is therefore economically large.

Interestingly, the impact of initial conditions on earnings trajectories is heterogeneous by parental income group, which is a novel finding and one of this paper's main contributions. Those in the bottom 20% of the parental income distribution have larger earnings declines compared to when the unemployment rate at graduation is 1 p.p. higher, with their earnings declining by on average \$2200 real CAD per year for the first 3 years after graduation. Meanwhile, for individuals with parental income in the top 80% of the parental income distribution, labor earnings decline by on average only \$1500 real CAD per year for the first three years after graduation. Thereafter, both groups face similar earnings declines when the initial unemployment rate is lower, regardless of their ranking in the parental income distribution as shown in Table A1 in the Appendix.

Thus, individuals from the lowest socio-economic class are more vulnerable to initial labor market conditions, namely the unemployment rate at graduation, compared to their peers from middle and upper class backgrounds. This relationship is not purely mechanical, since I am measuring the response to labor earnings that do not include direct transfers from parents to children. My findings hold even when conditioning on observables including gender, degree type, and major grouping. I use additional robustness

checks to condition for the top 3 and top 10 institutions as well as living at the parental home, and the results are similar.

These findings are consistent with parents acting as a financial safety net. Individuals from higher and middle income families are further from the borrowing constraint, and are more able to select jobs with higher earnings in the first three years after graduation. In comparison, those individuals from low income families lack the liquidity to do so, and are more pressed to take whichever job is available as they are closer to the borrowing constraint, leading to on average lower earnings. During an economic downturn, particularly the 2008-09 Great Recession, employment becomes harder to find, further amplifying the impact of liquidity constraints and labor market conditions and widening differences by parental income group.

The average earnings of both groups eventually recover to their long run trend. Those from the bottom 20% of the parental income distribution recover slightly faster in approximately nine years after graduation, while those in the top 80% of the parental income distribution take on average ten to eleven years after graduation to return to the long-run trend. This is in line with the observation that those from the bottom 20% of the parental income distribution have a lower average trend of labor earnings compared with individuals from wealthier parents, and therefore their average earnings catch up slightly faster.

Looking at individuals with positive labor earnings only who are in the workforce, we can analyze the effect of initial labor market conditions on log earnings, $\log y_{ctpi}$. Table A2 in the Appendix documents the results. When conditioning on individual positive earnings, individuals with worse initial labor market conditions likewise have a significant and persistent decrease in their earnings. On average, I find that individuals who graduate and enter the labor market with a 1 percentage point higher unemployment rate have 3 percent lower earnings each year for six years following graduation. This persistent result is robust to controls for age, gender, being in school, degree and institution type as well as major and year fixed effects.

Once again, initial labor market conditions have a different impact on earnings by parental income group. Similarly to the results shown in levels, individuals in the bottom 20% of the parental income distribution are more affected by the initial unemployment rate compared to their peers in the top 80%. This heterogeneous effect is concentrated in the first two years after graduation, after which both groups experience similar declines in earnings. Workers from the poorest 20% families have on average approximately 5.5% lower earnings in the year of graduation, and 3.5% lower earnings one year after graduation, all else equal. For workers from the middle and top of the family income distribution, earnings decrease less relative to their trend, with earnings falling on average only 2.5% in the graduation year and 1.5% in the year after graduation when conditioning for the same observables including age, gender, degree type, major group and two digit NAICS industry of employment.

To approximate full-time workers, I focus on individuals whose annual earnings exceed the real full-time minimum wage, which is approximately CAD \$10,000 in 2002 constant dollars. Consistent with earlier findings, earnings for these proxied full-time workers are significantly influenced by initial labor market conditions and vary across parental income groups. The results are shown in Table A2. A one percentage point increase in the unemployment rate at the time of graduation is associated with an average annual decline in labor earnings of 2% per year over the first seven years post-graduation, holding other factors constant. These effects are both highly persistent and economically significant, underscoring the substantial scarring impact of adverse labor market conditions at graduation on long-term earnings.

The effects on earnings vary significantly by parental income group. Individuals from the bottom quintile of the parental income distribution experience larger earnings declines relative to trend compared to their peers from middle- and upper-income backgrounds. Earnings for those in the bottom 20% of the parental income distribution decline by approximately 3.5%, while the corresponding decline for those in the top 80% is only 1.5%. These differences remain robust when controlling for observable characteristics as shown

in Table A3, including gender, degree, major, age, and broad industry grouping at the two-digit NAICS level, ensuring comparability across individuals. Notably, this differential effect persists for up to three years after graduation, after which both groups experience similarly lowered earnings for seven years after graduation when entering a worse initial labor market.

Together these results underscore that individuals from different parental income groups experience different effects of initial labor market conditions. The early post-graduation years, which are critical for career development, reveal significantly different responses to the initial unemployment rate by parental income group, particularly during the first three years following graduation. Workers from low-income families are particularly vulnerable to adverse initial labor market conditions, as they often lack a parental safety net and are more likely to face borrowing constraints. Consequently, they are more inclined to accept available employment out of financial necessity, and are more likely to sort into low earnings jobs, despite having similar observables to their peers from more affluent parents.

Together these empirical results reinforce the importance of parental income in explaining earnings differences across groups. In the following section, I develop a structural model to better understand the relationship between parental income and labor earnings dynamics.

3. Model

Section 3 lays out the structure of the model. The model is a continuous-time heterogeneous agent model with incomplete markets and search frictions, in the spirit of Krusell-Mukoyama-Sahin. I model parental income as a “safety-net” for new labor market entrants, where parental income determines initial assets a_0^p . Individuals face borrowing constraints which increase the reliance on initial parental wealth and lead to different values of unemployment by assets. Individuals are also endowed with human capital level $h \in \mathcal{H}$, that affects their earnings and employment outcomes. Human capital is time invariant in the model, and we can think of human capital as post-secondary schooling level. At a given instant, each individual is either unemployed (U) or employed (E), with frictional transitions across labor market sectors $s \in \{U, E\}$. The sectors differ in terms of earnings, as described below. In addition, agents choose the optimal levels of consumption c , and savings a' . The state variables are current assets a , human capital h , current labor market sector s and the aggregate state of the continuous-time economy $\mathcal{Z}$.

3.1. Primitives

Consumer Preferences

The economy has a measure one of consumers, who have CRRA utility with relative risk aversion γ

$$U(c) = \frac{1}{1-\gamma} c^{1-\gamma} \quad \gamma > 1, \gamma \neq 1$$

$$U(c) = \log(c) \quad \gamma = 1$$

where c denotes the consumption level.

Individual labor income differs by labor market sector s , as well as by human capital level h and the aggregate state of the economy $\mathcal{Z}$, similarly to Huckfeldt (2022). Labor supply is assumed to be inelastic. Workers have standard time-additive preferences with discount

factor ρ and do not value leisure. There are two labor market sectors: unemployment (U) and employment (E). In unemployment, workers receive income from home production, b^U . Employed workers are paid a wage $\omega(a, h, t, \mathcal{Z})$ which depends on their current asset level a , their human capital h , and aggregate economic conditions $\mathcal{Z}$. This wage schedule is continuously negotiated with the firm via Nash bargaining, as will be described in further detail in section ? below.

Production

There are a large measure of potential firms in the economy, one for each job. A firm without a worker can post a single vacancy at cost ξ . An active firm employs one worker with human capital h and rents capital per worker k_t at each instant t .

Firms produce output $y = \mathcal{Z}F(k, h)$ which depends on aggregate economic conditions $\mathcal{Z}$, capital per worker k , and the worker's human capital h . Production F is increasing and concave in human capital, and increasing and strictly concave in physical capital per worker. In the quantification, I assume the functional form $F(k, h) = h + k^\alpha$. Output is either consumed, invested or used for vacancy creation, as elaborated below.

3.2. Asset Market

There is only one asset in this economy, capital. Firms rent a stock of capital per worker, k_t , continuously at interest rate r_t and capital depreciates at rate δ_k . The gross real interest rate is given by $R_t = 1 + r_t - \delta_k$. Consumers face a borrowing constraint in this asset market, which in the quantification is assumed to be a no-borrowing condition. That is, consumers can only save at the market rate r_t to smooth consumption. In equilibrium, firms demand capital, and consumers supply capital through their savings.

3.3. Labor Market: Search and Matching

Unemployed workers with mass u_t and vacant firms v_t meet randomly at every instant according to the constant returns to scale Diamond-Mortensen-Pissarides aggregate match-

ing function:

$$m(u_t, v_t; \mathcal{Z}) = \chi u_t(\mathcal{Z})^\zeta v_t(\mathcal{Z})^{(1-\zeta)}$$

where $u_t(\mathcal{Z})$ denotes the total number of unemployed searchers looking for a job and $v_t(\mathcal{Z})$ is the total number of vacancies posted by vacant firms at instant t .

The job-finding rate p_t and vacancy-filling rate q_t are as follows, and vary endogenously with aggregate productivity $\mathcal{Z}$. The job-finding rate depends on the mass of unemployed workers in the economy, u_t , and the vacancy-filling rate depends on the number of vacant job openings v_t .

$$p_t(\mathcal{Z}) = \frac{m(u_t, v_t; \mathcal{Z})}{u_t(\mathcal{Z})}, \quad q_t(\mathcal{Z}) = \frac{m(u_t, v_t; \mathcal{Z})}{v_t(\mathcal{Z})}$$

Both the job-finding and vacancy-filling rates can be expressed as a function of the endogenous labor market tightness θ_t , given by $\theta(\mathcal{Z}) = \frac{v(\mathcal{Z})}{u(\mathcal{Z})}$.

$$p_t(\theta_t, \mathcal{Z}) = m(1, \theta_t; \mathcal{Z}), \quad q_t(\theta_t, \mathcal{Z}) = m(\theta_t^{-1}, 1; \mathcal{Z})$$

Job separations, on the other hand, are exogenous, with an employed worker transitioning to unemployment at constant Poisson rate η .

From the above assumptions, unemployment u_t evolves according to the following expression:

$$\dot{u}_t = \eta \times (1 - u_t) - p_t(\mathcal{Z}) \times u_t$$

3.4. Worker's Problem

Workers can either be employed or unemployed. The following HJB equation characterizes the worker's value function in unemployment. At every instant, an unemployed worker chooses an optimal level of consumption taking into account the flow utility and the effect

of saving or dissaving, $\dot{a}$, on their value of unemployment. At rate $p_t(\mathcal{Z})$, the unemployed worker meets a vacancy, in which case their value function changes as they transition to employment. The individual takes into account the effect of the sectoral transition, as well as how the value function of unemployment evolves with time captured in the last term below, when optimally choosing their current consumption.

$$\begin{aligned} \rho V^U(a, h; t, \mathcal{Z}) = \max_c \left\{ u(c) + \frac{\partial V^U(a, h; t, \mathcal{Z})}{\partial a} \underbrace{((r_t - \delta^k)a + b^U - c)}_{\dot{a}} \right. \\ \left. + p_t(\mathcal{Z}) (V^E(a, h; t, \mathcal{Z}) - V^U(a, h; t, \mathcal{Z})) \right. \\ \left. + \frac{\partial V^U(a, h; t, \mathcal{Z})}{\partial t} \right\} \end{aligned}$$

Consumers face a borrowing constraint which creates a lower bound on their assets, $\underline{a}$. The following state-constraint boundary condition summarizes the borrowing constraint for unemployed workers.

$$\partial_a V^U(\underline{a}, h; t, \mathcal{Z}) \geq u'(b^U + (r_t - \delta^k)\underline{a})$$

Similarly, the below HBJ equation describes the worker's value function when employed. Once again, the worker chooses consumption optimally taking into consideration the flow utility payoff, the effect of saving and accumulating assets on their value function, the evolution of the value function over time, as well as the evolution of the value function with exogenous separation at rate η - where the worker transitions from employment to unemployment. A key difference between the value function of employed and unemployed workers are earnings. When employed, a worker receives earnings in the form of the wage $\omega(a, h; t, \mathcal{Z})$, whereas when unemployed, their earnings are their home production/unemployment benefit b^U . This difference in earnings is highlighted in the savings/dissavings term $\dot{a}$ below.

$$\begin{aligned} \rho V^E(a, h; t, \mathcal{Z}) = \max_c \left\{ u(c) + \frac{\partial V^E(a, h; t, \mathcal{Z})}{\partial a} \underbrace{((r_t - \delta^k)a + \omega(a, h; t, \mathcal{Z}) - c)}_{\dot{a}} \right. \\ \left. + \eta \left(V^U(a, h; t, \mathcal{Z}) - V^E(a, h; t, \mathcal{Z}) \right) \right. \\ \left. + \frac{\partial V^E(a, h; t, \mathcal{Z})}{\partial t} \right\} \end{aligned}$$

At the borrowing constraint, employed workers have assets denoted by $\underline{a}$ and face the following state-constraint boundary condition:

$$\partial_a V^E(\underline{a}, h; t, \mathcal{Z}) \geq u'(\omega(\underline{a}, h; t, \mathcal{Z}) + (r_t - \delta^k)\underline{a})$$

Note that the boundary condition is similar for employed and unemployed workers, with the key difference being that earnings for employed workers come from the wage $\omega(a, h; t, \mathcal{Z})$, while unemployed workers have earnings b^U from home production.

Consumption and Savings Policy Functions

Taking the first order conditions of the worker HJB equations yields workers' policy functions for consumption and savings, denoted by $c(a, h, s; t, \mathcal{Z})$ and $\dot{a}(a, h, s; t, \mathcal{Z})$ respectively for consumers in sector $s \in \{U, E\}$.

We can write the consumption policy function generally for both employed and unemployed workers as follows:

$$c(a, h, s; t, \mathcal{Z}) = (u')^{-1}(\partial_a V(a, h, s; t, \mathcal{Z}))$$

The savings policy function is nearly identical for employed and unemployed workers, with slight differences in the source of earnings as we have seen above. Employed workers earn wage $\omega(a, h; t, \mathcal{Z})$, while unemployed workers receive home production b^U . Thus we

can express the savings policy function as follows:

$$\dot{a}(a, h, s = E; t, \mathcal{Z}) = \omega(a, h; t, \mathcal{Z}) + (r_t - \delta^k)a - c(a, h, s = E; t, \mathcal{Z})$$

$$\dot{a}(a, h, s = U; t, \mathcal{Z}) = b^U + (r_t - \delta^k)a - c(a, h, s = U; t, \mathcal{Z})$$

3.5. Firm's Problem

Firms create jobs, pay wages to employed workers, rent capital from consumers, and produce. As discussed above, output depends on aggregate conditions $\mathcal{Z}$ and is increasing human and physical capital in the firm's production function $F(h, k)$. Wages are negotiated with workers at every instant via Nash bargaining, and the firm optimally chooses the amount of physical capital per worker, k , to rent at every instant according to the real interest rate r_t . Firms discount future profits at the market rate $r_t - \delta^k$. As in the standard Diamond-Mortensen-Pissarides framework, each firm is a single job that can employ at most one worker, and vacant firms can post a single vacancy. Here J denotes the value of a firm employing one worker, and the value of a vacant firm with no workers is denoted by J_0 .

The firm's HJB value function $J(a, h; t, \mathcal{Z})$ when employing a worker with assets a and human capital h is as follows.

$$\begin{aligned} (r_t - \delta^k)J(a, h; t, \mathcal{Z}) = \max_k \bigg\{ & \mathcal{Z}F(h, k) - \omega(a, h, t, \mathcal{Z}) - r_t \times k \\ & + \frac{\partial J(a, h; t, \mathcal{Z})}{\partial a} \dot{a}(a, h, s = E; t, \mathcal{Z}) \\ & + \eta(J_0 - J(a, h; t, \mathcal{Z})) \\ & + \frac{\partial J(a, h; t, \mathcal{Z})}{\partial t} \bigg\} \end{aligned}$$

The firm optimally chooses the amount of capital per worker k to rent at every instant,

taking into account the effect of k on the firm's profits as well as its continuation value. The value of the firm also depends on the worker's saving choice $\dot{a}$, which the firm takes as given, since it changes the worker's assets and therefore affects future wage bargaining and subsequent wages. The firm also takes into account the possibility of becoming exogenously separated from its worker at Poisson rate η and becoming a vacant firm, as well as how the value of the firm evolves over time.

A potential vacant firm takes into account the distribution of agents across states of the world, given by $g(a, h, s, t, \mathcal{Z})$ with sector $s \in \{U, E\}$, and decides whether or not to enter. Upon entering, a vacant firm J_0 pays a the flow cost ξ every instant t to post a vacancy and create a job.

The value of the vacant firm J_0 is as follows:

$$(r_t - \delta^k)J_0(t, \mathcal{Z}) = -\xi + q_t(\mathcal{Z}) \int_a \int_h J(a, h; t, \mathcal{Z}) \frac{g(a, h, u; t, \mathcal{Z})}{u_t} dh da + \partial_t J_0(t, \mathcal{Z})$$

The first term captures the payment of the flow vacancy posting cost ξ , and the second term denotes the expected value of matching and producing with an unemployed worker of assets a and human capital h . At rate $q_t(\mathcal{Z})$, which varies endogenously with the labor market tightness and aggregate economic conditions, posted vacancies are filled and the firm produces following $J(a, h; t, \mathcal{Z})$. When deciding to enter, the potential firm considers the pool of unemployed workers with which it could randomly match, since the worker's human capital h and assets a affect the firm's output and costs (via wages). The expectation of worker type across the unemployment pool is summarized in the marginal distribution $\frac{g(a, h, u; t, \mathcal{Z})}{u_t}$ of unemployed workers over assets and human capital. The value of the vacant firm also evolves over time t , as shown in the last term.

With free entry of firms, firms will post vacancies until the value of any potential entrant equals to zero and we have the following equilibrium condition:

$$\xi = q_t(\mathcal{Z}) \int_a \int_h J(a, h; t, \mathcal{Z}) \frac{g(a, h, u; t, \mathcal{Z})}{u_t} dh da$$

Namely, in equilibrium with free entry, the flow vacancy posting cost ξ is equal to the expected value of meeting an unemployed worker with assets a and human capital h and filling the vacancy.

The matched firm's first-order condition with respect to k implies

$$r_t = \mathcal{L}F_k(k, h)$$

as in the usual relationship equating the marginal cost of capital to the marginal factor demand for capital per worker.

3.6. Wage Setting

A newly formed job-worker match generates a surplus that must be split between the worker and the firm. As is common in the DMP literature, wages are determined by Nash bargaining between a matched worker and firm. I assume that the firm cannot commit to future wages, and therefore flow wages are bargained at every instant, similarly to Krusell et. al. (2010). Current assets are taken into account in the bargaining because they affect worker and thereby firm's continuation values. The Nash bargaining solves the following problem with maximizer $w = \omega(a, h, \mathcal{Z})$. Here $\nu \in (0, 1)$ denotes the worker's bargaining power. For notational simplicity, I omit the time subscripts.

$$\max_w \left(V^E(w, a, h, \mathcal{Z}) - V^U(a, h, \mathcal{Z}) \right)^\nu \left(J(w, a, h, \mathcal{Z}) - J_0(a, h, \mathcal{Z}) \right)^{1-\nu}$$

The wage depends on worker's asset and human capital level, as well as aggregate economic conditions. Current assets are taken into account in the bargaining because they affect worker and thereby firm's continuation values. This dependence is highlighted by the dependency of V^E and J on w in the above equation. In continuous time, V^E and J are both linear in $\omega(a, h, \mathcal{Z})$, which makes solving the problem computationally more tractable.

3.7. Stationary Equilibrium

3.7.1. Definition of Stationary Equilibrium

The following equations describes a stationary equilibrium. To simplify notation, I will omit the time subscripts, the aggregate economic state $\mathcal{Z}$ and partial derivatives. Recall that the individual states are assets a , human capital h , and sector $s \in \{U, E\}$.

The stationary recursive equilibrium consists of

- a set of value functions for consumers and firms $\{V^U(a, h), V^E(a, h), J(a, h), J_0\}$;
- a set of policy functions $\{c(a, h, s), \dot{a}(a, h, s)\}$;
- a distribution over assets, human capital, and sectors $g(a, h, s)$;
- initial asset, human capital and sector distributions, a_0 , h_0 and s_0
- a set of prices $\{r, \omega(a, h)\}$;
- a capital stock per worker k ; and
- labor market tightness θ

such that:

- (i) Consumers are optimizing. Given prices and the labor market tightness, the policy functions $\{c(a, h, s), \dot{a}(a, h, s)\}$ solve the consumer optimization problem, with the value functions $V^U(a, h)$ and $V^E(a, h)$ for unemployed and employed workers respectively.
- (ii) The stationary distribution $g(a, h, s)$ is consistent with the distribution of workers.
- (iii) Firms are optimizing. Given prices, labor market tightness, and the stationary distribution, k optimally solves the firm's problem with corresponding value functions $J(a, h)$ and J_0 .
- (iv) There is free entry of firms. Given prices, labor market tightness, and the stationary

distribution, the values of a vacancy adjusts such that J_0 satisfies free entry.

- (v) In the labor market, the job-finding and vacancy-filling rates are given by the labor market tightness θ . The measure of vacancies is given by $v = u \times \theta$ where the unemployment rate u is

$$u = \frac{\eta}{\eta + p(\theta)} = \int_a \int_h g(a, h, u) dh da$$

- (vi) Wages are negotiated at every instant via generalized Nash bargaining between a matched worker and firm.
- (vii) The asset market clears. The aggregate supply of capital is determined from consumers' savings, while the aggregate demand of capital by firms is K . For symmetric firms, aggregate capital can be expressed in relation to all employed workers' capital per worker $K = (1 - u)k$.
- (viii) The goods market clears. Firms and home production produce the final good, and output is used for consumption, investment in physical capital, and vacancy creation.

3.8. Calibration

To quantitatively solve the model, I assign standard values for the assigned parameters in Table 1. The risk aversion parameter in the CRRA utility function is set with $\gamma = 1$. Aggregate economic conditions $\mathcal{Z}$ are normalized to one in the stationary baseline. In the baseline, the borrowing limit is set to zero to impose a no-borrowing constraint. The borrowing constraint can be relaxed to allow a fixed level of borrowing $\underline{a}$ for all agents.

TABLE 1. Assigned Parameters

Parameter	Description	Value
γ	Risk aversion	1
$\underline{a}$	Borrowing limit	0
$\mathcal{Z}$	Aggregate economic conditions, baseline	1
ζ	Elasticity of matching wrt unemployment	0.72
ν	Worker's bargaining power	0.72
α	Capital share in production	0.3

Following Shimer (2005), the elasticity of the matching function with respect to unemployment is set to $\zeta = 0.72$, and the bargaining parameter ν which captures the worker's bargaining power is set according to the Hosios condition $\nu = \zeta = 0.72$. In the standard DMP framework, the Hosios condition would ensure efficiency. However, with workers' precautionary savings, efficiency is not guaranteed in this setting.

For the remaining parameters shown in table 2, I jointly estimate the parameter set ψ_p by matching moments. The estimated values minimize the following objective function following Freund (2024):

$$\mathcal{G}(\psi_p) = \sum_{j=1}^7 \left(\frac{\hat{m}_j - m_j(\psi_p)}{\frac{1}{2}\|\hat{m}_j\| + \frac{1}{2}\|m_j(\psi_p)\|} \right)^2$$

where $\hat{m}_j$ refers to the empirical moment and $m_j(\psi_p)$ denotes its model counterpart.

I use standard values for the discount factor ρ , targeting the annual real rate of return of capital as in Krusell et. al. (2010). The depreciation rate of capital is calibrated according

to the investment-output ratio.

TABLE 2. Calibrated Parameters

Parameter	Description	Value
δ_k	Depreciation rate of capital	0.022
ρ	Discount factor	0.01
b^u	Home production	0.75
χ	Matching efficiency	1.793
η	Exogenous separation rate	0.1038
h_2/h_1	Human capital productivity ratio	4.00
ξ	Vacancy posting cost	0.464

TABLE 3. Model vs Data Moments

Moment Name	Model	Data	Source
Investment-output ratio	0.188	0.200	Krusell et. al. (2010)
Ann. real rate of return on capital	0.039	0.040	Krusell et. al. (2010)
Log ratio top/bottom 20% assets	1.678	1.674	Data
Ratio top/bottom 20% assets	5.356	5.333	Data
Home production	$0.365\bar{w}$	$0.4\bar{w}$	Shimer (2005)
Monthly job finding rate	0.453	0.45	Shimer (2005)
Monthly separation rate	0.034	0.034	Shimer (2005)
Skill wage ratio	1.155	1.20	Data
Labor market tightness	1.032	1	Shimer (2005)

Following the literature from Shimer (2005), the value of home production is estimated to match 40% of the average wage when employed. The matching efficiency χ governs the job-finding rate, as increasing the efficiency in the matching function increases the rate of matches between unemployed workers and vacancies. The exogenous separation rate, on the other hand, is directly disciplined by the monthly separation rate.

The observed skill wage premium in the microdata disciplines the relative efficiency of human capital in production. In the quantification, human capital takes two levels. The lower level of human capital, h_1 , corresponds to individuals with less than a bachelor's degree, while the higher level of human capital, h_2 , parallels individuals with a bachelor's degree or higher. This two-tier classification follows standard practice in the labor

literature, reflecting the sizable earnings gap between bachelor's and above graduates compared to those with some college, such as associate degrees or certificates. Bringing the model to the data, I target the approximately 20% college wage premium. This data moment is in line with aggregate Canadian data on the relative earnings premium of bachelor and higher graduates to post-secondary graduates who have attained less than a bachelor's degree.

Finally, as in Shimer (2005), the labor market tightness target is normalized to one, and the flow vacancy-posting cost ξ adjusts to achieve this target.

3.9. Simulations

Using the calibrated model, I conduct several quantitative exercises. In the first exercise, I simulate a contraction in aggregate conditions $\mathcal{Z}$ to emulate the Great Recession's impact on initial labor market conditions. The model allows me to then decompose these effects by initial human capital and by initial asset levels.

As in the data, I study the effect of initial labor market conditions on the earnings trajectories of post-secondary graduates. With the calibrated model, I simulate two steady-state economies. The first is a baseline economy with favorable initial labor market conditions, corresponding to a stationary environment where $\mathcal{Z}$ is normalized to one. As in the empirical section, this baseline economy will be denoted in the graph by a solid blue line. The second economy represents a stationary environment with adverse initial labor market conditions, where aggregate productivity $\mathcal{Z}$ declines by 3% relative to the baseline. This second counterfactual economy emulates graduating into a labor market with a high unemployment rate such as during the Great Recession in Canada, and is plotted as an orange dashed line. I simulate a large mass of agents in each economy by parental income group, comparing individuals from low-income families (bottom 20% of the parental income distribution) to those from high-income families in the top 20% of parental income. Average earnings after graduation are plotted in Figure 2 below. Consistent with the data, few graduates are employed full-time at graduation, so I initialize all individuals

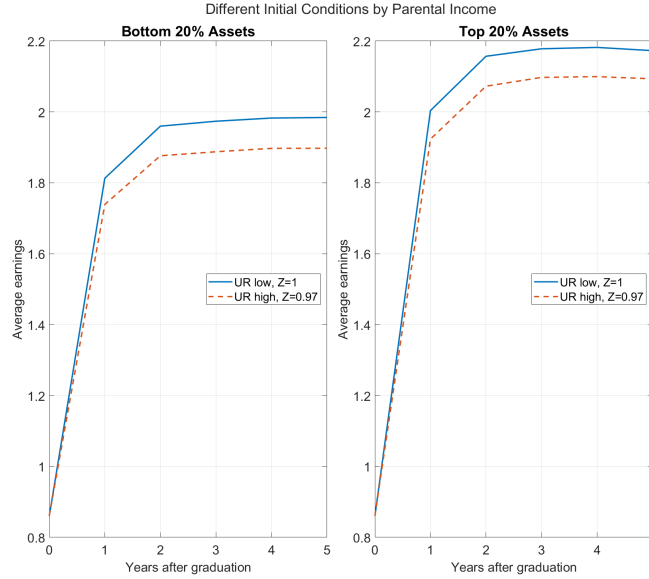


FIGURE 2. Model Moments, Average Earnings by Initial Labor Market Conditions and Parental Income

as unemployed. Over time, unemployed graduates meet vacancies and transition to employment, which raises average earnings quickly in early years. This mirrors the rise in average earnings after graduation pattern seen in the cohort level data in Figure 1.

When the initial labor market conditions are worse (shown in the orange dashed line) the job-finding rates decline and individuals from all parental backgrounds take longer to find employment. Moreover, recessions lower match surplus, so equilibrium wages fall relative to the baseline.

Second, wages are on average lower for individuals from low-income backgrounds, and with a smaller parental safety net, in any state of the world. Since individuals from low-income parents have very little safety net, they find themselves closer to the borrowing constraint. The nonlinearities near the borrowing limit cause graduates from low-income backgrounds to have much lower values of unemployment, and reduces their bargaining power substantially when they do match with a firm. Consequently, their initial earnings are lower than those of peers from more affluent families.

Both these patterns reflect what we see in the data. Figure 3 shows the equivalent data

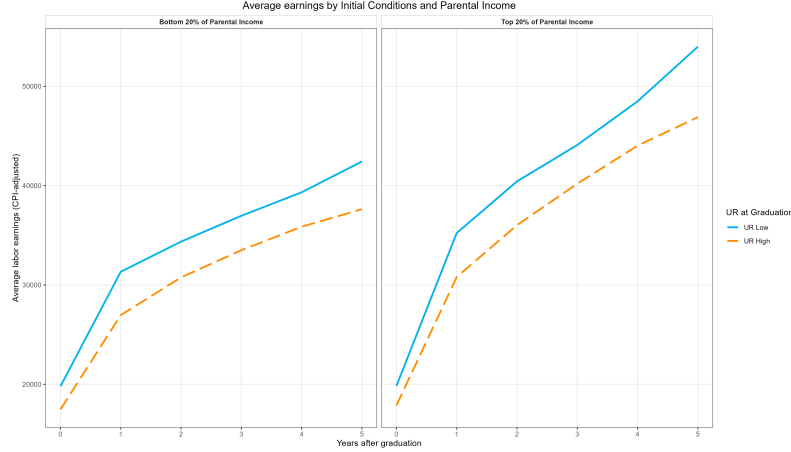


FIGURE 3. Aggregate Data Moments, Average Earnings by Initial Conditions & Parental Income

analogue to the simulations, plotting average earnings by parental income group under baseline and adverse labor market conditions. The model does a good job of qualitatively matching the initial wage patterns by parental income and initial labor market conditions, and thus provides qualitative support for the parental-safety net mechanism. The present exercise intentionally uses steady states to isolate the mechanism; transitional dynamics will be quantified by introducing an unexpected MIT shock, which delivers the wage mean reversion observed after recessions. Likewise, time invariant human capital is a deliberate simplification to sharpen the role of borrowing constraints and search. Allowing for human capital to accumulate through on the job earnings is a natural extension to capture lifetime earnings profiles.

Moreover, in the model, human capital is time invariant, which abstracts from on the job learning that can further increase workers' productivity and earnings. Thus we cannot model lifetime earnings rising in the model. Since the main focus of the paper is in the first years following graduation, where time variation in human capital is limited, this stationary exercise serves as a clean and informative benchmark for evaluation the model mechanism against the data.

4. Conclusion

This paper studies the unequal impact of initial labor market conditions by parental income, and how it can create persistent differences in long-run outcomes. Using large Canadian administrative tax data, I analyze the relationship between parental income and child labor market trajectories, focusing on how aggregate economic conditions influence the wage dynamics of new labor market entrants. I find significant and lasting effects on wages that are unequal by parental income group, with entrants from lower-income families being more vulnerable to downturns and experiencing larger earnings declines when starting their careers during a recession. Motivated by these insights, I develop a model incorporating financial constraints and frictional sector choices within a heterogeneous agent model to quantify the role of initial parental income.

The mechanism I explore is that parental income acts as a safety net, whereby individuals with low-income parents find themselves closer to the borrowing constraint compared to their peers with better-off parents. Individuals with lower-income parents are initially more likely to be in jobs with lower earnings, compared to their peers from higher family income backgrounds. When the initial labor market conditions worsen such as during a recession, this mechanism is amplified as the job finding rate decreases.

My findings help to better understand a key driver of intergenerational inequality and have important policy implications. I suggest that targeted economic policy of unemployment insurance for new labor market entrants can alleviate scarring effects for marginal workers from lower-income families. Such policies are particularly relevant in contexts where inter-generational income and human capital mobility is low.

References

- Achdou, Yves, Jiequn Han, Jean-Michel Lasry, Pierre-Louis Lions, and Benjamin Moll. 2022. "Income and wealth distribution in macroeconomics: A continuous-time approach." *The Review of Economic Studies* 89 (1): 45–86.
- Alon, Titan, Matthias Doepke, Jane Olmstead-Rumsey, and Michèle Tertilt. 2020. "The Impact of COVID-19 on Gender Equality." NBER Working Paper No. 26947, National Bureau of Economic Research.
- Altonji, Joseph G, Lisa B Kahn, and Jamin D Speer. 2016. "Cashier or consultant? Entry labor market conditions, field of study, and career success." *Journal of Labor Economics* 34 (S1): S361–S401.
- Ameriks, John, Joseph Briggs, Andrew Caplin, Matthew D. Shapiro, and Christopher Tonetti. 2016. "The Long-Term-Care Insurance Puzzle: Modeling and Measurement." NBER Working Paper No. 22726, National Bureau of Economic Research.
- Bardóczy, Bence. "Labor-Market Matching with Precautionary Savings." Technical note discussing Krusell, Mukoyama, and Sahin (2010).
- Bardóczy, Bence. 2022. "Spousal Insurance and the Amplification of Business Cycles." PhD dissertation, Northwestern University.
- Boar, Corina. 2021. "Dynastic Precautionary Savings." *The Review of Economic Studies* 88 (6): 2735–2765.
- Chadwick, Laura, and Gary Solon. 2002. "Intergenerational Income Mobility Among Daughters." *American Economic Review* 92 (1): 335–344.
- Chetty, Raj, John Friedman, Emmanuel Saez, Nick Turner, and Danny Yagan. 2017. "Mobility Report Cards: The Role of Colleges in Intergenerational Mobility." NBER Working Paper No. 23618, National Bureau of Economic Research.
- Chetty, Raj, Nathaniel Hendren, Patrick Kline, and Emmanuel Saez. 2014. "Where Is the Land of Opportunity? The Geography of Intergenerational Mobility in the United States." *The Quarterly Journal of Economics* 129 (4): 1553–1623.
- Choi, HwaJung, Kathleen McGarry, and Robert F. Schoeni. 2016. "Consumption and the Extended Family." *Economics Letters* 140: 34–38.
- Corak, Miles. 2013. "Income Inequality, Equality of Opportunity, and Intergenerational Mobility." *Journal of Economic Perspectives* 27 (3): 79–102.
- Corak, Miles, and Patrizio Piraino. 2011. "The Intergenerational Transmission of Employers." *Journal of Labor Economics* 29 (1): 37–68.

- Cox, Donald. 1990. "Intergenerational Transfers and Liquidity Constraints." *The Quarterly Journal of Economics* 105: 187–217.
- Genda, Yuji, Ayako Kondo, and Souichi Ohta. 2010. "Long-Term Effects of a Recession at Labor Market Entry in Japan and the United States." *Journal of Human Resources* 45 (1): 157–196.
- Hampf, Franziska, Marc Piopiunik, and Simon Wiederhold. 2020. "The Effects of Graduating from High School in a Recession: College Investments, Skill Formation, and Labor Market Outcomes." *CESuFi Working Paper No. 8252*.
- Hershbein, Brad. 2012. "Graduating High School in a Recession: Work, Education, and Home Production." *The B.E. Journal of Economic Analysis Policy* 12 (1): 1–32.
- Huckfeldt, Christopher. 2022. "Understanding the Scarring Effect of Recessions." *American Economic Review* 112 (4): 1273–1310.
- Jovanovic, Boyan. 1979. "Job Matching and the Theory of Turnover." *Journal of Political Economy* 87 (5): 972–990.
- Kahn, Lisa B. 2010. "The Long-Term Labor Market Consequences of Graduating from College in a Bad Economy." *Labor Economics* 17 (2): 303–316.
- Kaila, Martti, Emily Nix, and Krista Riukula. 2024. "The impact of an early career shock on intergenerational mobility." *Journal of Labor Economics* (forthcoming).
- Kaplan, Greg. 2012. "Moving Back Home: Insurance against Labor Market Risk." *Journal of Political Economy* 120 (3): 446–512.
- Kaplan, Greg, and Giovanni L Violante. 2014. "A model of the consumption response to fiscal stimulus payments." *Econometrica* 82 (4): 1199–1239.
- Kaplan, Greg, Giovanni L Violante, and Justin Weidner. 2014. "The wealthy hand-to-mouth." National Bureau of Economic Research.
- Kondo, Ayako. 2015. "Differential Effects of Graduating during a Recession across Gender and Race." *Journal of Labor Economics* 4 (1): 23.
- Krusell, Per, Toshihiko Mukoyama, and Aysegul Sahin. 2010. "Labour-market matching with precautionary savings and aggregate fluctuations." *The Review of Economic Studies* 77 (4): 1477–1507.
- Ljungqvist, Lars, and Thomas J. Sargent. 1998. "The European Unemployment Dilemma." *Journal of Political Economy* 106 (3): 514–550.
- McGarry, Kathleen. 2016. "Dynamic Aspects of Family Transfers." *Journal of Public Economics* 137: 1–13.
- Oreopoulos, Philip, Till von Wachter, and Andrew Heisz. 2012. "The Short- and Long-Term Career Effects of Graduating in a Recession." *American Economic Journal: Applied Economics* 4 (1): 1–29.

- Postel-Vinay, Fabien, and Jean-Marc Robin. 2002. "Equilibrium Wage Dispersion with Worker and Employer Heterogeneity." *Econometrica* 70 (6): 2295–2350.
- Shimer, Robert. 2005. "The cyclical behavior of equilibrium unemployment and vacancies." *American Economic Review* 95 (1): 25–49.
- Speer, Jamin. 2016. "Wages, Hours, and the School-to-Work Transition: The Consequences of Leaving School in a Recession for Less-Educated Men." *The B.E. Journal of Economic Analysis Policy* 16 (1): 97–124.
- Stinson, Martha, and Christopher Wignall. 2018. "Fathers, Children, and the Intergenerational Transmission of Employers." Working Paper 18-12, Center for Economic Studies, U.S. Census Bureau.
- Torche, Florencia. 2011. "Is a College Degree Still the Great Equalizer? Intergenerational Mobility across Levels of Schooling in the United States." *American Journal of Sociology* 117 (3): 763–807.
- Wachter, Till Von. 2020. "The persistent effects of initial labor market conditions for young adults and their sources." *Journal of Economic Perspectives* 34 (4): 168–194.

Appendix A. Appendix



FIGURE A1. Unemployment Rate in Canada, 1975- 2020
Recessions shaded in gray

TABLE A1. Real Labor Earnings Response to 1 p.p. increase in initial unemployment rate UR_{c0} , bottom 20% and top 80% of parental income

Dependent variable: Real Labor Earnings in 2002 CAD

	(1)	(2)	(3)	(4)	(5)	(6)
UR Results / Years After Graduation						
Reference Group is Bottom 20% of Parental Income						
$\gamma_{0,j=0}$	-2,478.416*** (269.735)	-2,564.204*** (266.568)	-2,533.158*** (264.923)	-2,614.899*** (262.212)	-2,616.457*** (262.215)	-2,601.437*** (256.766)
$\gamma_{0,j=1}$	-2,358.815*** (273.03)	-2,406.333*** (269.825)	-2,511.994*** (268.16)	-2,660.625*** (265.415)	-2,662.264*** (265.417)	-2,614.033*** (259.896)
$\gamma_{0,j=2}$	-2,151.935*** (273.512)	-2,155.812*** (270.301)	-2,365.000*** (268.636)	-2,566.850*** (265.886)	-2,568.559*** (265.889)	-2,407.400*** (260.355)
$\gamma_{0,j=3}$	-2,211.117*** (338.807)	-2,135.527*** (334.83)	-2,128.101*** (332.763)	-2,549.252*** (329.363)	-2,551.209*** (329.366)	-2,273.418*** (322.506)
$\gamma_{0,j=5}$	-2,337.990*** (399.288)	-2,143.261*** (394.602)	-1,854.507*** (392.171)	-2,522.854*** (388.177)	-2,524.605*** (388.179)	-2,127.655*** (380.096)
$\gamma_{0,j=8}$	-1,645.904*** (468.911)	-1,570.259*** (463.406)	-1,522.383*** (460.545)	-2,178.521*** (455.843)	-2,180.230*** (455.845)	-1,929.869*** (446.349)
$\gamma_{0,j=10}$	175.217 (485.019)	243.156 (479.325)	56.43 (476.368)	-588.499 (471.5)	-590.29 (471.502)	-361.678 (461.681)
UR x Family Income Group Results						
Interaction of UR_{c0} x Family Income Group p : Results for Top 80% of Parental Income						
$\gamma_{p,j=0}$	490.642* (277.879)	483.262* (274.617)	540.715** (272.922)	546.344** (270.128)	545.698** (270.128)	440.186* (264.503)
$\gamma_{p,j=1}$	910.644*** (284.312)	901.228*** (280.975)	695.978** (279.244)	692.424** (276.384)	691.784** (276.385)	640.292** (270.628)
$\gamma_{p,j=2}$	1,010.040*** (286.629)	1,001.086*** (283.264)	828.098*** (281.518)	819.889*** (278.635)	819.264*** (278.636)	752.918*** (272.832)
$\gamma_{p,j=3}$	676.469* (357.404)	717.560** (353.209)	555.255 (351.03)	600.813* (347.436)	600.115* (347.436)	617.670* (340.199)
$\gamma_{p,j=5}$	-23.355 (423.72)	28.508 (418.746)	-22.738 (416.161)	136.114 (411.9)	135.288 (411.901)	216.641 (403.319)
$\gamma_{p,j=8}$	-609.917 (498.199)	-547.585 (492.351)	-567.683 (489.311)	-264.921 (484.3)	-265.829 (484.3)	-183.363 (474.21)
$\gamma_{p,j=10}$	-1,274.259** (515.017)	-1,222.922** (508.971)	-1,261.372** (505.829)	-945.286* (500.648)	-946.083* (500.648)	-846.190* (490.217)
Age	Y	Y	Y	Y	Y	Y
Refyear FE	Y	Y	Y	Y	Y	Y
Sex		Y	Y	Y	Y	Y
In School			Y	Y	Y	Y
Degree Type				Y	Y	Y
College or Uni					Y	Y
Major						Y
R^2	0.147	0.167	0.177	0.194	0.194	0.227
Adjusted R^2	0.147	0.167	0.177	0.194	0.194	0.227

Note: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

TABLE A2. Real Log Positive Labor Earnings Response to 1 p.p. increase in initial UR_{c0} , bottom 20% and top 80% of parental income

<i>Dependent variable: Real Positive Log Labor Earnings</i>							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
UR Results j Years After Graduation							
Reference Group is Bottom 20% of Parental Income							
$\gamma_{0,j=0}$	-0.057*** (0.009)	-0.058*** (0.009)	-0.057*** (0.008)	-0.060*** (0.008)	-0.060*** (0.008)	-0.061*** (0.008)	-0.053*** (0.008)
$\gamma_{0,j=1}$	-0.031*** (0.009)	-0.032*** (0.009)	-0.036*** (0.008)	-0.040*** (0.008)	-0.040*** (0.008)	-0.039*** (0.008)	-0.033*** (0.008)
$\gamma_{0,j=2}$	0.001 (0.009)	0.001 (0.009)	-0.007 (0.009)	-0.012 (0.008)	-0.012 (0.008)	-0.007 (0.008)	0.001 (0.008)
$\gamma_{0,j=3}$	-0.025** (0.011)	-0.022** (0.011)	-0.020* (0.011)	-0.031*** (0.01)	-0.031*** (0.01)	-0.022** (0.01)	-0.012 (0.01)
$\gamma_{0,j=5}$	-0.034*** (0.013)	-0.028** (0.013)	-0.016 (0.013)	-0.033*** (0.012)	-0.033*** (0.012)	-0.019 (0.012)	-0.009 (0.012)
$\gamma_{0,j=8}$	-0.013 (0.015)	-0.011 (0.015)	-0.009 (0.015)	-0.028* (0.015)	-0.028* (0.015)	-0.02 (0.014)	-0.011 (0.014)
$\gamma_{0,j=10}$	0.029* (0.016)	0.031** (0.016)	0.024 (0.015)	0.004 (0.015)	0.004 (0.015)	0.01 (0.015)	0.018 (0.014)
UR x Family Income Group Results							
Interaction of UR_{c0} x Family Income Group p: Results for Top 80% of Parental Income							
$\gamma_{p,j=0}$	0.026*** (0.009)	0.025*** (0.009)	0.028*** (0.009)	0.027*** (0.009)	0.027*** (0.009)	0.022*** (0.008)	0.020** (0.008)
$\gamma_{p,j=1}$	0.032*** (0.009)	0.031*** (0.009)	0.021** (0.009)	0.019** (0.009)	0.019** (0.009)	0.017** (0.008)	0.019** (0.008)
$\gamma_{p,j=2}$	0.020** (0.009)	0.020** (0.009)	0.011 (0.009)	0.01 (0.009)	0.009 (0.009)	0.006 (0.009)	0.007 (0.008)
$\gamma_{p,j=3}$	0.020* (0.011)	0.021* (0.011)	0.012 (0.011)	0.013 (0.011)	0.013 (0.011)	0.012 (0.011)	0.014 (0.01)
$\gamma_{p,j=5}$	-0.018 (0.014)	-0.016 (0.013)	-0.019 (0.013)	-0.014 (0.013)	-0.014 (0.013)	-0.012 (0.013)	-0.01 (0.012)
$\gamma_{p,j=8}$	-0.024 (0.016)	-0.021 (0.016)	-0.022 (0.016)	-0.012 (0.015)	-0.012 (0.015)	-0.01 (0.015)	-0.008 (0.014)
$\gamma_{p,j=10}$	-0.026 (0.017)	-0.025 (0.016)	-0.028* (0.016)	-0.017 (0.016)	-0.017 (0.016)	-0.013 (0.016)	-0.009 (0.015)
Age	Y	Y	Y	Y	Y	Y	Y
Refyear FE	Y	Y	Y	Y	Y	Y	Y
Sex		Y	Y	Y	Y	Y	Y
In School			Y	Y	Y	Y	Y
Degree Type				Y	Y	Y	Y
College or Uni					Y	Y	Y
Major						Y	Y
Industry FE Two-Digit NAICS							Y
R^2	0.184	0.198	0.22	0.238	0.238	0.276	0.331
Adjusted R^2	0.184	0.198	0.22	0.238	0.238	0.276	0.331

Note: * p<0.1; ** p<0.05; *** p<0.01

TABLE A3. Real Log Positive Full-Time Labor Earnings Response to 1 p.p. increase in initial UR_{c0} , bottom 20% and top 80% of parental income

<i>Dependent variable: Real Positive Log Labor Earnings above 10k</i>							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
UR Results j Years After Graduation							
Reference Group is Bottom 20% of Parental Income							
$\gamma_{0,j=0}$	-0.034*** (0.006)	-0.034*** (0.006)	-0.034*** (0.006)	-0.035*** (0.006)	-0.036*** (0.006)	-0.035*** (0.006)	-0.032*** (0.005)
$\gamma_{0,j=1}$	-0.040*** (0.005)	-0.040*** (0.005)	-0.040*** (0.005)	-0.041*** (0.005)	-0.042*** (0.005)	-0.039*** (0.005)	-0.033*** (0.005)
$\gamma_{0,j=2}$	-0.039*** (0.005)	-0.038*** (0.005)	-0.040*** (0.005)	-0.044*** (0.005)	-0.045*** (0.005)	-0.040*** (0.005)	-0.030*** (0.005)
$\gamma_{0,j=3}$	-0.034*** (0.006)	-0.032*** (0.006)	-0.031*** (0.006)	-0.041*** (0.006)	-0.042*** (0.006)	-0.036*** (0.006)	-0.026*** (0.006)
$\gamma_{0,j=5}$	-0.030*** (0.007)	-0.026*** (0.007)	-0.022*** (0.007)	-0.038*** (0.007)	-0.038*** (0.007)	-0.030*** (0.007)	-0.023*** (0.007)
$\gamma_{0,j=8}$	-0.013 (0.015)	-0.011 (0.015)	-0.009 (0.015)	-0.028* (0.015)	-0.028* (0.015)	-0.02 (0.014)	-0.011 (0.014)
$\gamma_{0,j=10}$	-0.003 (0.009)	-0.001 (0.009)	-0.003 (0.009)	-0.019** (0.009)	-0.019** (0.009)	-0.012 (0.008)	-0.003 (0.008)
UR x Family Income Group Results							
Interaction of UR_{c0} x Family Income Group p : Results for Top 80% of Parental Income							
$\gamma_{p,j=0}$	0.012* (0.006)	0.011* (0.006)	0.012* (0.006)	0.014** (0.006)	0.014** (0.006)	0.015** (0.006)	0.017*** (0.006)
$\gamma_{p,j=1}$	0.015*** (0.006)	0.016*** (0.005)	0.012** (0.005)	0.012** (0.005)	0.012** (0.005)	0.012** (0.005)	0.013*** (0.005)
$\gamma_{p,j=2}$	0.017*** (0.006)	0.016*** (0.005)	0.014*** (0.005)	0.014*** (0.005)	0.014*** (0.005)	0.014*** (0.005)	0.012** (0.005)
$\gamma_{p,j=3}$	0.006 (0.007)	0.007 (0.007)	0.005 (0.007)	0.005 (0.006)	0.005 (0.006)	0.008 (0.006)	0.007 (0.006)
$\gamma_{p,j=5}$	-0.007 (0.008)	-0.007 (0.008)	-0.007 (0.008)	-0.005 (0.007)	-0.005 (0.007)	-0.002 (0.007)	-0.001 (0.007)
$\gamma_{p,j=8}$	-0.021** (0.009)	-0.020** (0.009)	-0.020** (0.009)	-0.013 (0.009)	-0.013 (0.009)	-0.01 (0.008)	-0.011 (0.008)
$\gamma_{p,j=10}$	-0.021** (0.010)	-0.020** (0.009)	-0.021** (0.009)	-0.013 (0.009)	-0.013 (0.009)	-0.011 (0.009)	-0.011 (0.008)
Age	Y	Y	Y	Y	Y	Y	Y
Refyear FE	Y	Y	Y	Y	Y	Y	Y
Sex		Y	Y	Y	Y	Y	Y
In School			Y	Y	Y	Y	Y
Degree Type				Y	Y	Y	Y
College or Uni					Y	Y	Y
Major						Y	Y
Industry FE Two-Digit NAICS							Y
R^2	0.184	0.198	0.22	0.238	0.238	0.276	0.331
Adjusted R^2	0.184	0.198	0.22	0.238	0.238	0.276	0.331

Note: * p<0.1; ** p<0.05; *** p<0.01